Invoice and PO review packets

The system in one sentence

Explain invoice differences and give an AP analyst a clear review packet.

Key design calls

Start with review drafts. Rejected: Automatic action. Why: Staff retain every decision.

Use supplied sources. Rejected: Open browsing. Why: Claims need checked evidence.

Code checks hard facts. Rejected: Model authority. Why: Values and access must be repeatable.

Who acts and who decides

Action	Autonomy
Check supplied evidence	acts alone
Draft an explanation	suggests
Send or take action	human decides

Top risks by RPN

FM4 | Revoked source access | RPN 108

FM1 | Currency differs | RPN 96

FM2 | Ambiguous OCR digit | RPN 84

FM3 | Wrong supplier PO | RPN 84

FM6 | Stale concurrent replacement | RPN 84

Launch blockers and owners

H1, H3, H5: Read supplied invoices and read-only...; One workday response target, coordinator...; Proposed quality target is 95%... Owner: AP operations lead.

H2: Authorized staff supply source bundles... Owner: Integration owner.

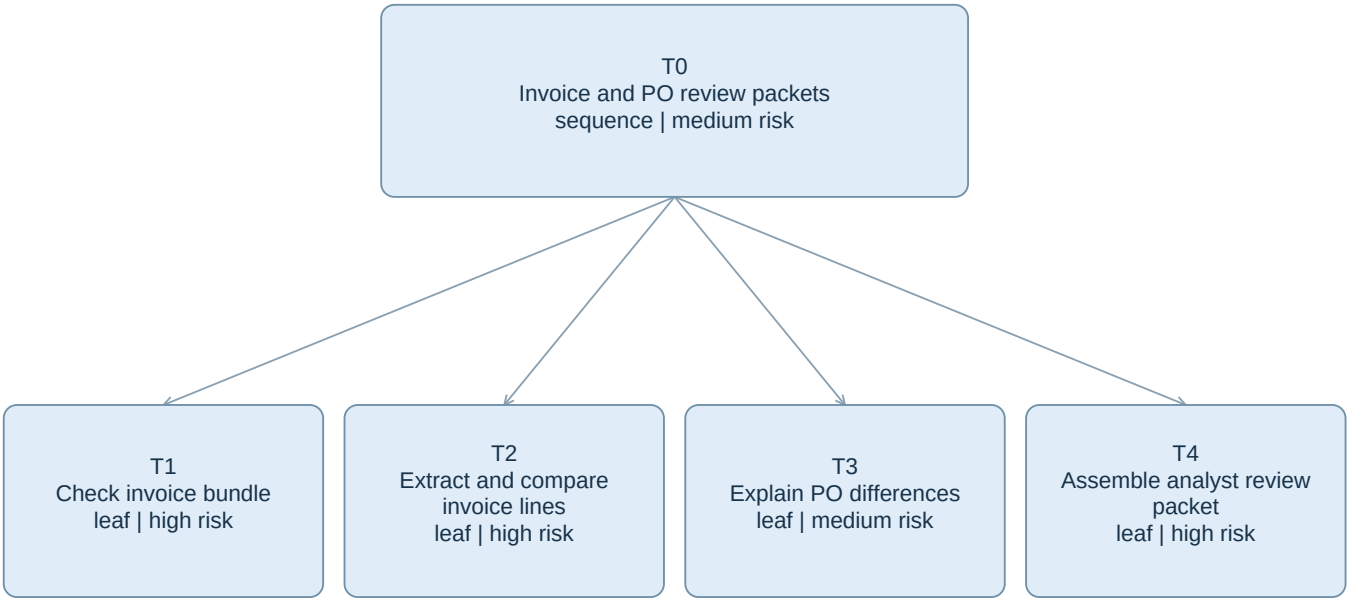
Illustrative cost per case

USD 0.00720 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: independent-agent; round 1; needs-work.

Proposed design. Full rules and blockers: design.json and assumptions.md.

Task tree



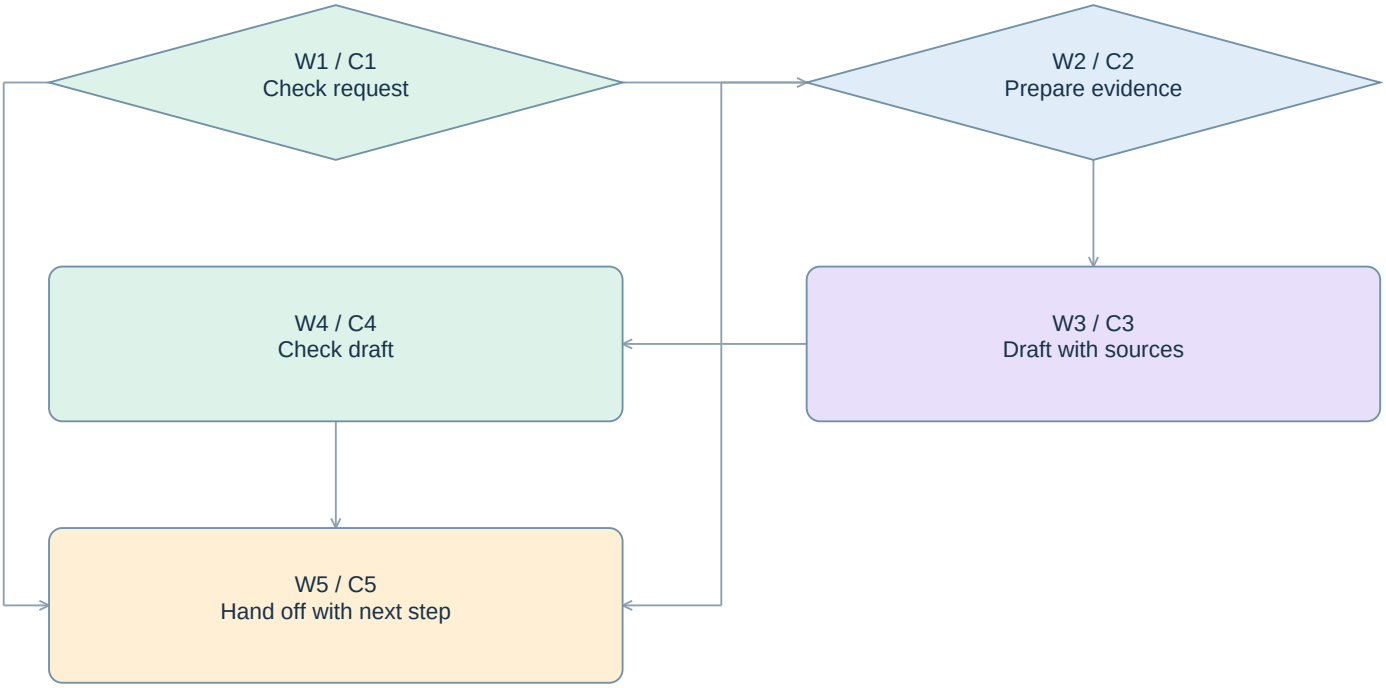
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

Read supplied invoices and read-only PO snapshots. Explain line differences for analyst review. Exclude posting, payment, supplier changes, tax advice and approval of invoices.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

Workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

Top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	-	-	-	-	-	-	-	-	-
9	-	-	FM4	-	-	-	-	-	-	-
8	-	-	FM1	-	-	-	-	-	-	-
7	-	-	FM2, FM3, FM6, FM7, FM9, FM10, FM11, FM12, FM13	-	-	-	-	-	-	-
6	-	-	FM5, FM8	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 Currency differs	Code prohibits cross-currency comparisons; analyst obtains a correct PO.
FM2 Ambiguous OCR digit	No guessed numeric fields; null extraction blocks computed comparison.
FM3 Wrong supplier PO	Trusted scope matches invoice and PO before document contents reach model.
FM4 Revoked source access	Fresh access checks before source reads and final ready output.
FM5 Model output unavailable	Bounded call and explicit fallback; no ready success claim.
FM6 Stale concurrent replacement	Compare revision at intake and handoff; same requestId/revision returns same artifact; changed intent needs next revision.
FM7 Missing receipt	C4 verifies receipt before ready. A failed receipt remains unconfirmed; staff may create a new local export. No external commit exists.
FM8 Cancellation	Cancellation precedes draft; cancellation during a call discards response and returns cancelled receipt.
FM9 Overdue human review	Packet includes proposed one-workday target, owner and backup. An operator checks pending list at daily opening; overdue remains visibly pending. Receiving...
FM10 Exact duplicate request	Return same local artifact for exact request/version; changed revision never reuses a draft.
FM11 Conflicting supplied evidence	Do not pick one conflicting record silently; name both sources and ask staff to resolve.
FM12 Missing supplied evidence	Allow provisional intake, list missing source and owned next step; no invented claim.
FM13 Forged citation	Verify citation ID/version/quote against frozen authorized view, never only syntax.

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Code: validation	Check case envelope	T1
C2 / Code: tools	Build trusted evidence view	T2
C3 / Agents (LLM)	Draft bounded recommendation	T3
C4 / Code: validation	Check claims and output proof	T3, T4
C5 / Code: human triggers	Prepare human handoff	T4

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

Architecture and model budget

F1 - Workflow choice

Choice: A small deterministic state machine with one bounded drafting call.

Rejected: Free tool-using agent; multi-agent chain.

Trade-off: Few states and no operational writes favor a simple flow; more agents add disagreement and latency.

F2 - Permissions and data

Choice: Read-only supplied evidence and local artifact export; no business write credentials.

Rejected: Connect full operational accounts.

Trade-off: A wider connection adds exposure without helping this review-only release.

Access: Separate source read principal from model; source allowlist enforced before retrieval; minimize, redact, expire per approved policy.

F3 - Storage and deployment options

Choice: Choose encrypted existing team storage with access checks and a relational run manifest; verify environment before purchase.

Rejected: AWS object store plus relational ledger; GCP object store plus relational ledger; Azure object store plus relational ledger; self-hosted object...

Trade-off: Compare same criteria: scoped reads, region, retention, audit, operations effort and price. Existing team storage wins provisionally; self-hosting adds operational burden. Retrieval...

Access: Separate source read principal from model; source allowlist enforced before retrieval; minimize, redact, expire per approved policy.

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

Model cost

Illustrative model cost is USD 0.00720 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

Evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	EV9	EV10	EV14	EV15
FM1	-	X	-	-	-	-	-	-	-	-	-	-
FM2	-	-	X	-	-	-	-	-	-	-	-	-
FM3	-	-	-	X	-	-	-	-	-	-	-	-
FM4	-	-	-	-	X	-	-	-	-	-	-	-
FM5	-	-	-	-	-	X	-	-	-	-	-	-
FM6	-	-	-	-	-	-	X	-	-	-	-	-
FM7	-	-	-	-	-	-	-	X	-	-	-	-
FM8	-	-	-	-	-	-	-	-	X	-	-	-
FM9	-	-	-	-	-	-	-	-	-	X	-	-
FM10	-	-	-	-	-	-	-	-	-	-	X	-
FM11	-	-	-	-	-	-	-	-	-	-	-	X
FM12	-	-	-	-	-	-	-	-	-	-	-	-
FM13	-	-	-	-	-	-	-	-	-	-	-	-
T1	X	X	X	X	X	-	X	-	X	-	X	-
T2	X	X	X	X	-	-	-	-	-	-	-	X
T4	X	X	X	X	X	X	X	X	X	X	X	X

Evaluation coverage matrix - continued

Risk / task	EV16	EV17
FM1	-	-
FM2	-	-
FM3	-	-
FM4	-	-
FM5	-	-
FM6	-	-
FM7	-	-
FM8	-	-
FM9	-	-
FM10	-	-
FM11	-	-
FM12	X	-
FM13	-	X
T1	X	-
T2	X	-
T4	X	X

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

Evaluation plan

Test	Layer	Purpose
EV1	golden	Useful bounded packet
EV2	golden	Currency differs
EV3	golden	Ambiguous OCR digit
EV4	golden	Wrong supplier PO
EV5	golden	Revoked source access
EV6	golden	Model output unavailable
EV7	golden	Stale concurrent replacement
EV8	golden	Missing receipt
EV9	golden	Cancellation
EV10	golden	Overdue human review
EV11	offline	Held-out staff usefulness and evidence audit
EV12	online	Review-only operating monitor
EV13	release	Release gate for all specified branches
EV14	golden	Exact duplicate request
EV15	golden	Conflicting supplied evidence
EV16	golden	Missing supplied evidence
EV17	golden	Forged citation

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

Measurement and rollout

R1 - Business measure

Measure: Analyst active minutes per invoice packet; weekly active review minutes / packets reviewed. Baseline unknown; proposed improvement goal set after baseline.

Signals: Fewer unsupported claims reduces rework; usefulness can lower review effort but must be measured.; No private-data leakage is a safety guardrail, not a claimed...

R2 - Shadow on historical cases

Phase / owner: shadow / AP operations lead

Gate: Entry: approved scope and lawful sample access. Exit: all release goldens and held-out citation/privacy checks pass; domain owner signs evidence.

R3 - Assist-only pilot

Phase / owner: assist pilot / AP operations lead

Gate: Entry: shadow exit and H1-H3/H5 verified. Review two weekly cycles, staffed capacity confirmed. Exit: no safety failures and useful drafts >=H5; owner may extend review-only usage. Scale keeps human review and all...

R4 - Measure review effort

Test: Randomize eligible case families to manual or assistant draft; stratify routine/exception and operator. Same source family stays in one arm; hide treatment drafts from control. Collect baseline variance of active minutes, choose minimum useful difference with owner, compute...

Decision gate: Entry uses baseline/power plan, never already-achieved pilot results. Stop for any critical failure.

R5 - Accountable owner and stop plan

Phase / owner: all / AP operations lead

Gate: Lead owns backlog, privacy incident response and release decision; technical owner owns adapters and evidence receipts. Stop use on safety breach; preserve scoped audit then manual fallback.

These are reader summaries. Complete choices, alternatives, gates and owners are in PRD.md and design.json.

Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Build a review-only tool

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	3/4
D2	3/4	D8	3/4
D3	2/4	D9	2/4
D4	2/4	D10	2/4
D5	3/4	D11	3/4
D6	3/4	D12	2/4

IA1 major OPEN: C4 requires an artifact receipt before ready but C5 creates the local receipt after C4. Specify the exact upstream producer, persistence contract and binding...

IA2 major OPEN: Successful handoffInput only carries C4.checked metadata, while checked omits the draft text/claim citations and typed domain result. C5 output also lacks review content. Preserve...

IA3 major OPEN: EV5, EV7, EV9 and EV14 expect zero/additional-zero model calls yet mockSequence contains model -> source-grounded draft and passRule says verify ordered mocks. EV1/EV17 give...

IA4 major resolved: Resolved by final iteration 6 reader export: actual PDF page 6 now displays F1-F3 choices/rejected alternatives/tradeoffs/access; page 10 displays R1-R5 measures, phase owners, gates...

IA5 major OPEN: Forged citation test is not a prompt-injection test. Invoice/source documents are untrusted model input but no risk/control/eval exercises embedded instructions to hide a mismatch...

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.